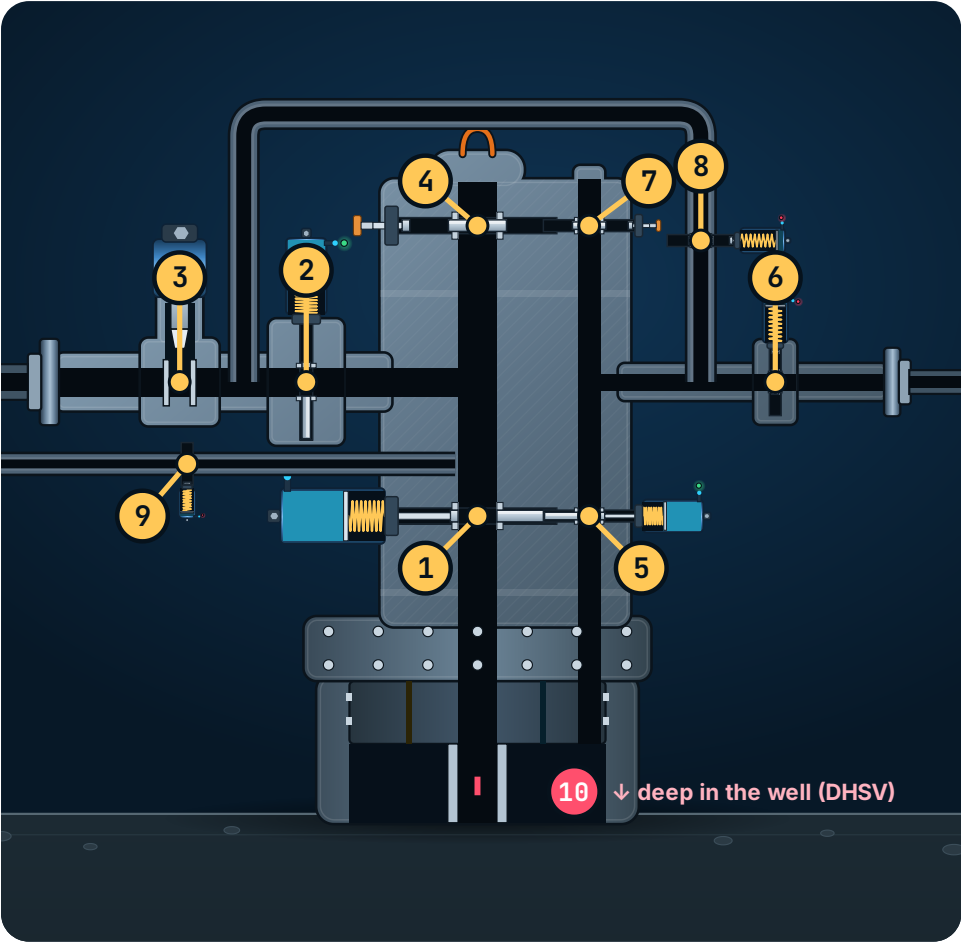


# Valve cheat sheet



| VALVE                       | WHAT IT DOES                                                             | NORMAL                 | NO PRESSURE |
|-----------------------------|--------------------------------------------------------------------------|------------------------|-------------|
| 1 PMV<br>Production master  | Main isolation of the well; part of the secondary barrier                | OPEN                   | Closes      |
| 2 PWV<br>Production wing    | Opens and closes the way to the flowline; first valve to close in an ESD | OPEN                   | Closes      |
| 3 CHOKE<br>Production choke | Sets how much flows (pressure drop); the insert is retrievable           | ADJUSTED               | -           |
| 4 PSV<br>Production swab    | Vertical access for wireline / intervention tools                        | CLOSED                 | Closes      |
| 5 AMV<br>Annulus master     | Access to the annulus (tubing-casing) for monitoring                     | OPEN                   | Closes      |
| 6 AWV<br>Annulus wing       | Side outlet from the annulus                                             | CLOSED                 | Closes      |
| 7 ASV<br>Annulus swab       | Vertical access to the annulus                                           | CLOSED                 | Closes      |
| 8 XOV<br>Crossover          | With the AMV open: bleeds annulus pressure into the flowline             | CLOSED                 | Closes      |
| 9 CIV<br>Chemical injection | Injects chemicals such as MEG to prevent hydrates                        | OPEN<br>when injecting | Closes      |
| 10 DHSV<br>Downhole safety  | Closes the tubing deep in the well – the last line of defence            | OPEN                   | Closes      |

### EMERGENCY SHUTDOWN – CLOSING ORDER

1 PWV → 2 PMV → 3 DHSV

The wing valve is designed to close against the flow. The **downhole safety valve** closes last – the last line of defence.

### FROM COMMAND TO MOTION

MCS → umbilical → SCM  
→ solenoid → DCV → actuator

Valves are **held open by hydraulic pressure** and **closed by springs**. Lose power or pressure → the springs close the valve (fail-safe). Sensors report back; an ROV can operate the valves directly.

### TWO INDEPENDENT, TESTED BARRIERS

**Primary:** tubing · packer · downhole safety valve  
**Secondary:** casing and cement · wellhead · tubing hanger · **tree with its valves**  
Barrier elements are tested regularly to prove they still work.

### QUIZ ANSWERS

1 · B – closes automatically when hydraulic pressure is lost  
2 · C – the downhole safety valve  
3 · B – the crossover valve (with the annulus master valve)